The empirical recurrence for OEIS A236814: recovering a conjecture that is not written down

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Abstract

OEIS A236814 records "Empirical recurrence of order 87", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 164$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236814 is "Number of (n+1)X(3+1) 0..2 arrays with the maximum plus the lower median minus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$1162, 10702, 98380, 956456, 9360398, 93804194, 947327860, 9682975968, \ \ldots$$

The entry states:

> Empirical recurrence of order 87 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:35:13, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 164$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 164.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 328$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{87} c_i\, a(n-i),$$

c_1	30	c_{23}	17804737602207351381	c_{45}	10759465608016029277005640 35	c_{67}
c_2	-50	c_{24}	-53494269584001636377	c_{46}	$-107021999653456840236357 5063$	c_{68}
c_3	-6914	c_{25}	-148703596391108404276	c_{47}	$-1793181179402408172139583460$	c_{69}
c_4	56825	c_{26}	665814537078362186550	c_{48}	2738321572065985791249514692	c_{70}
c_5	620000	c_{27}	849882989649911503256	c_{49}	2159655119906908266891347986	c_{71}
c_6	-8823890	c_{28}	-6376552886165955266520	c_{50}	$-5286501216701410928180483672$	c_{72}
c_7	-21231056	c_{29}	-2023985156801248518927	c_{51}	$-1435019391261447843334971234$	c_{73}
c_8	712649613	c_{30}	4820624489564776840 2378	c_{52}	7945484919659038793673460710	c_{74}
c_9	-686140706	c_{31}	-19247842124293965032943	c_{53}	$-750953383754591097918759582$	c_{75}
c_{10}	-36295497697	c_{32}	$-29015366453846741139920 0$	c_{54}	$-9336452242143762257216928131$	c_{76}
c_{11}	112914575365	c_{33}	29616280407397005112839 1	c_{55}	3732624406398569798036613577	c_{77}
c_{12}	1245544005845	c_{34}	138440834730765515647218 9	c_{56}	8450925237534172928356779326	c_{78}
c_{13}	-6347365002307	c_{35}	$-2299558632285427578387943$	c_{57}	$-5986149817181084701274288674$	c_{79}
c_{14}	-28888854584644	c_{36}	$-5119547531344472937270184$	c_{58}	$-5642029091504791094667419214$	c_{80}
c_{15}	225736251950572	c_{37}	12733260417312019822688593	c_{59}	6307064348820514803896774458	c_{81}
c_{16}	407765750489979	c_{38}	13726171282244713700698824	c_{60}	2446334246314087416900936389	c_{82}
c_{17}	-5738443521471890	c_{39}	$-54417116335348544879521595$	c_{61}	$-4828081534818868150920326786$	c_{83}
c_{18}	-1149890832378082	c_{40}	$-20362079076182437439941352$	c_{62}	$-295352484274418802990799580$	c_{84}
c_{19}	109370207550528170	c_{41}	184498821186955151470086115	c_{63}	273926889288016841585467327 2	c_{85}
c_{20}	-104043149918690319	c_{42}	$-251979551927951784764688 26$	c_{64}	$-479676332431650123540919378$	c_{86}
c_{21}	-1594501820813188222	c_{43}	$-4999803384914915199661125 78$	c_{65}	$-1136711462917031055123804740$	c_{87}
c_{22}	3101906979159742404	c_{44}	283252270672941367733545224	c_{66}	436355264688675694178205640	

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A236814.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.